

24K-0559

BAZIL UDDIN KHAN

BSCS-1H

Q1).

24k-0559 BAZIL UDDIN KHAN

CODE:

// 24k-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

**void Registerparticipants(int Totalelementsone, int
Totalelementstwo,string Eventone[],string Eventwo[]);**

**bool Attendancechecker(int Totalelementsone, int
Totalelementstwo,string Eventone[],string Eventwo[],string Name);**

**void TotalDonations(int Totalelementsone, int
Totalelementstwo,string Eventone[],string Eventwo[]);**

```
void Displaynames(int Totalelementsone, int Totalelementstwo,string  
Eventone[],string Eventtwo[]);
```

```
void Determinepopularity(int Totalelementsone, int  
Totalelementstwo,string Eventone[],string Eventtwo[]);
```

```
int main()
```

```
{
```

```
    int Totalelementsone;
```

```
    cout << " Enter total elements for event one : " <<endl;
```

```
    cin >> Totalelementsone;
```

```
    int Totalelementstwo;
```

```
    cout << " Enter total elements for event two : " <<endl;
```

```
    cin >> Totalelementstwo;
```

```
    if(Totalelementsone <= 5 && Totalelementstwo <= 5 &&  
Totalelementsone > 0 && Totalelementstwo > 0)
```

```
    {
```

```
        string Eventone[Totalelementsone];
```

```
        string Eventtwo[Totalelementstwo];
```

```
Registerparticipants( Totalelementsone, Totalelementstwo,
Eventone,Eventwo);

string Name;

cout << " Enter name : " << endl;

cin >> Name;

bool result = Attendancechecker( Totalelementsone,
Totalelementstwo, Eventone,Eventwo,Name);

cout << " The result of the search is : " << result << endl;

TotalDonations(Totalelementsone,
Totalelementstwo,Eventone,Eventwo);

Displaynames(Totalelementsone, Totalelementstwo, Eventone,
Eventwo);

Determinepopularity( Totalelementsone, Totalelementstwo,
Eventone, Eventwo);
}

else
{
    cout << " Retry you have crossed max limit " << endl;
}

return 0;
}
```

```
void Registerparticipants(int Totalelementsone, int
Totalelementstwo,string Eventone[],string Eventtwo[])
{
    cout << " Enter names for Event : "<< endl;

    for(int i =0; i < Totalelementsone;i++)
    {
        string Name;
        cout << " Enter names for event one " << endl;
        cin >> Name;
        Eventone[i] = Name;

    }

    for(int i =0; i < Totalelementstwo;i++)
    {
        string name;
        cout << " Enter names for event two " << endl;
        cin >> name;
        Eventtwo[i] = name;

    }
```

```
}
```

```
bool Attendancechecker(int Totalelementsone, int  
Totalelementstwo,string Eventone[],string Eventtwo[],string Name)
```

```
{
```

```
    bool found = false;
```

```
    for(int i =0;i < Totalelementsone; i++)
```

```
    {
```

```
        if(Eventone[i] == Name)
```

```
        {
```

```
            found = true;
```

```
        }
```

```
    }
```

```
    if(found == false)
```

```
    {
```

```
        for(int i =0;i < Totalelementsone; i++)
```

```
        {
```

```
            if(Eventtwo[i] == Name)
```

```
            {
```

```
                found = true;
```

```
    }  
    }  
  
    }  
    return found;  
}
```

```
void TotalDonations(int Totalelementsone, int  
Totalelementstwo,string Eventone[],string Eventtwo[])  
{  
    const int Donate = 10;  
    int Total =0;  
    for(int i =0; i < Totalelementsone;i++)  
    {  
        Total = Total + Donate;  
    }  
    for(int i =0; i < Totalelementstwo;i++)  
    {  
        Total = Total + Donate;  
    }  
}
```

```
cout << " Total donation done is " << Total << endl;
```

```
}
```

```
void Displaynames(int Totalelementsone, int Totalelementstwo,string  
Eventone[],string Eventtwo[])
```

```
{
```

```
    for(int i = Totalelementsone-1;i>=0;i--)
```

```
    {
```

```
        cout << " Name is " << Eventone[i] << endl;
```

```
    }
```

```
    for(int i = Totalelementstwo-1;i>=0;i--)
```

```
    {
```

```
        cout << " Name is " << Eventtwo[i] << endl;
```

```
    }
```

```
}
```

```
void Determinepopularity(int Totalelementsone, int  
Totalelementstwo,string Eventone[],string Eventtwo[])
```

```
{
```

```
cout << " Bar chart represents as " << endl;
```

```
cout << " Event one : " << endl;
```

```
for(int i =0; i < Totalelementsone;i++)
```

```
{
```

```
    cout << "      " << " * ";
```

```
}
```

```
cout << endl;
```

```
cout << " Eventtwo : " << endl;
```

```
for(int i =0; i < Totalelementstwo;i++)
```

```
{
```

```
    cout << "      " << " * ";
```

```
}
```

```
cout << endl;
```

```
}
```

OUTPUT:

main.cpp

Share

Run

```
1 // 24k-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5
6 void Registerparticipants(int Totalelementsone, int Totalelementstwo,string Eventone[]
    ,string Eventtwo[]);
7
8 bool Attendancechecker(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[],string Name);
9
10 void TotalDonations(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[]);
11
12 void Displaynames(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[]);
13
14
15 void Determinepopularity(int Totalelementsone, int Totalelementstwo,string Eventone[]
    ,string Eventtwo[]);
16
17
18 int main()
19 {
20
21     int Totalelementsone;
22     cout << " Enter total elements for event one : " <<endl;
23     cin >> Totalelementsone;
24     int Totalelementstwo;
25     cout << " Enter total elements for event two : " <<endl;
26     cin >> Totalelementstwo;
27
28     if(Totalelementsone <= 5 && Totalelementstwo <= 5 && Totalelementsone > 0 &&
```

Output

Enter total elements for event one :
4
Enter total elements for event two :
4
Enter names for Event :
Enter names for event one
WE
Enter names for event one
ERT
Enter names for event one
YOU
Enter names for event one
IOP
Enter names for event two
QWE
Enter names for event two
GHK
Enter names for event two
FR
Enter names for event two
JKL
Enter name :
JKL
The result of the search is : 1
Total donation done is 80
Name is IOP
Name is YOU
Name is ERT
Name is WE
Name is JKL
Name is FR
Name is GHK
Name is QWE

main.cpp

Share

Run

```
1 // 24k-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5
6 void Registerparticipants(int Totalelementsone, int Totalelementstwo,string Eventone[]
    ,string Eventtwo[]);
7
8 bool Attendancechecker(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[],string Name);
9
10 void TotalDonations(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[]);
11
12 void Displaynames(int Totalelementsone, int Totalelementstwo,string Eventone[],string
    Eventtwo[]);
13
14
15 void Determinepopularity(int Totalelementsone, int Totalelementstwo,string Eventone[]
    ,string Eventtwo[]);
16
17
18 int main()
19 {
20
21     int Totalelementsone;
22     cout << " Enter total elements for event one : " <<endl;
23     cin >> Totalelementsone;
24     int Totalelementstwo;
25     cout << " Enter total elements for event two : " <<endl;
26     cin >> Totalelementstwo;
27
28     if(Totalelementsone <= 5 && Totalelementstwo <= 5 && Totalelementsone > 0 &&
```

Output

Enter names for event one
WE
Enter names for event one
ERT
Enter names for event one
YOU
Enter names for event one
IOP
Enter names for event two
QWE
Enter names for event two
GHK
Enter names for event two
FR
Enter names for event two
JKL
Enter name :
JKL
The result of the search is : 1
Total donation done is 80
Name is IOP
Name is YOU
Name is ERT
Name is WE
Name is JKL
Name is FR
Name is GHK
Name is QWE
Bar chart represents as
Event one :
* * * *
Eventtwo :
* * * *

Q2).

24K0559 BAZIL-UDDIN-KHAN

CODE:

//24K0559 BAZIL-UDDIN-KHAN

```
#include <iostream>
```

```
using namespace std;
```

```
void AddValues(int Aqr [][][7],float Firstaverage[]);
```

```
void Pollutiondays(int Aqr [][][7]);
```

```
void Display(int Aqr [][][7]);
```

```
void MonthlyAqr(int aqr [][][28],float Firstaverage[],float  
lastAverage[],int Aqr [][][7]);
```

```
void Displayreport(int aqr [][][28],float Firstaverage[],float  
lastAverage[]);
```

```
int main()
```

```
{
```

```

cout << "Welcome to check on air quality index " << endl;

int Aqr[4][7]= {
    {120, 140, 160, 180, 200, 220, 240},
    {80, 100, 120, 140, 160, 180, 200},
    {60, 70, 80, 90, 100, 110, 120},
    {200, 220, 240, 260, 280, 300, 320}
};

float Firstaverage[4];
float lastAverage[4];
AddValues(Aqr,Firstaverage);
Pollutiondays(Aqr);
Display(Aqr);

int aqr[4][28] = {
    {120, 140, 160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220,
    240, 120, 140, 160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220,
    240}, // City 1
    {80, 100, 120, 140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200,
    80, 100, 120, 140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200}, //
    City 2
    {60, 70, 80, 90, 100, 110, 120, 60, 70, 80, 90, 100, 110, 120, 60, 70,
    80, 90, 100, 110, 120, 60, 70, 80, 90, 100, 110, 120}, // City 3
    {200, 220, 240, 260, 280, 300, 320, 200, 220, 240, 260, 280, 300,
    320, 200, 220, 240, 260, 280, 300, 320, 200, 220, 240, 260, 280, 300,
    320} // City 4

```

```
};
```

```
;
```

```
MonthlyAqr(aqr,Firstaverage,lastAverage,Aqr);
```

```
Displayreport( aqr , Firstaverage, lastAverage);
```

```
return 0;
```

```
}
```

```
void AddValues(int Aqr [][][7],float Firstaverage[])
```

```
{
```

```
for(int i =0; i<4;i++)
```

```
{
```

```
int Sum =0;
```

```
for(int j =0; j<7;j++)
```

```
{
```

```
Sum = Sum + Aqr[i][j];
```

```
}
```

```

float Average = (Sum/7);
Firstaverage[i] = Average;
cout << "Average for city " <<i+1 << " is " << Average;
cout << endl;
}

```

```

int HighestAqr = Firstaverage[0];
int index =0;
for(int i =0; i< 4;i++)
{
    if(Firstaverage[i] > HighestAqr)
    {
        HighestAqr = Firstaverage[i];
        index = i;
    }

}

cout << "Highest Aqr for city "<<index+1 << " that is " <<
HighestAqr << endl;
}

```

```

void Pollutiondays(int Aqr [][][7])

```

```
{  
    for(int i =0; i < 4;i++)  
    {  
        for(int j =0; j<7;j++)  
        {  
            if(Aqr[i][j] > 150)  
            {  
                cout << " Day is " << j+1 << " City is " << i+1 << endl;  
                cout << endl;  
            }  
        }  
    }  
}
```

```
void Display(int Aqr[][7])  
{  
    for(int i = 0; i < 4; i++)  
    {  
        cout << " City " << i+1 <<endl;  
  
        for(int j =0; j< 7; j++)  
        {
```

```
int increment = 0;
int aqr = Aqr[i][j];
bool flag = true;
while(flag)
{
    if(increment <= aqr )
    {
        cout << "*" ;
        cout << " ";
        increment = increment + 50;
    }
    else
    {
        flag = false;
    }

}

}

cout << endl;
}
}
```

```
void MonthlyAqr(int aqr[][28],float Firstaverage[],float lastAverage[],  
int Aqr[][7])
```

```
{
```

```
    for(int i =0; i < 4;i++)
```

```
    {
```

```
        for(int y =0; y < 7;y++ )
```

```
        {
```

```
            aqr[i][y] = Aqr[i][y];
```

```
        }
```

```
    }
```

```
float AVerage[4];
```

```
for(int i =0; i<4;i++)
```

```
{
```

```
    int Sum =0;
```

```
    for(int j =0; j<28;j++)
```

```
    {
```

```
        Sum = Sum + aqr[i][j];
```

```
    }
```

```
    float Average = (Sum/28);
```



```
Average[i] = Average;
```

```
}
```

```
for(int i =0; i<4;i++)
```

```
{
```

```
    int Sum =0;
```

```
    for(int j =21; j<28;j++)
```

```
    {
```

```
        Sum = Sum + aqr[i][j];
```

```
    }
```

```
    float Average = (Sum/7);
```

```
    lastAverage[i] = Average;
```

```
}
```

```
float Maxchange[4];
```

```
for(int i =0; i < 4;i++)
```

```
{
```

```
    if(Firstaverage[i] >= lastAverage[i])
```

```
    {
```

```

        Maxchange[i] = Firstaverage[i] - lastAverage[i];
    }
    else
    {
        Maxchange[i] = lastAverage[i] - Firstaverage[i];
    }
}
float HighestArq = Maxchange[0];
int index =0;

for(int i =0; i <4;i++)
{
    if(HighestArq > Maxchange[i])
    {
        HighestArq = Maxchange[i];
        index = i;
    }
}

cout << "Highest Aqr for city " <<index +1<< " that is " <<
HighestArq << endl;

```

```
}
```

```
void Displayreport(int aqr [][][28],float Firstaverage[],float  
lastAverage[])
```

```
{
```

```
    float Secondweek[4];
```

```
    float Thirdweek[4];
```

```
    for(int i =0; i<4;i++)
```

```
    {
```

```
        int sum =0;
```

```
        for(int j =7; j <14;j++)
```

```
        {
```

```
            sum = sum + aqr[i][j];
```

```
        }
```

```
        float avg = (sum/7);
```

```
        Secondweek[i] = avg;
```

```
    }
```

```
    for(int i =0; i<4;i++)
```

```
    {
```

```
        int sum =0;
```

```
        for(int j =14; j <21;j++)
```

```
        {
```

```

        sum = sum + aqr[i][j];
    }
    Thirdweek[i] = (sum/7);
}

int HighestAqr = 0;
int Index =0;
int lowestAqr = aqr[0][0];

for(int i =0; i<4;i++)
{
    for(int j =0; j<28;j++)
    {
        if(aqr[i][j] > HighestAqr)
        {
            HighestAqr = aqr[i][j];
            Index = i;
        }
    }
}

int index =0;
for(int i =0; i<4;i++)
{

```

```
for(int j =0; j<28;j++)
{
    if(aqr[i][j] < lowestAqr)
    {
        lowestAqr = aqr[i][j];
        index = i;
    }
}

cout << "Highest Aqr for city in " << Index+1<< " is " << HighestAqr
<< endl;
```

```
cout << "lowest Aqr for city in " << index+1<< " is " << lowestAqr <<
endl;
```

```
for(int i =0; i <4; i++)
{
    cout << " City first week avg " << i+1 << " Average is " <<
Firstaverage[i] << endl;

    cout << " City second week avg " << i+1 << " Average is " <<
Secondweek[i] << endl;

    cout << " City third week avg " << i+1 << " Average is " <<
Thirdweek[i] << endl;
```

```

        cout << " City last week avg " << i+1 << " Average is " <<
lastAverage[i] << endl;

    }
}

```

OUTPUT:

main.cpp	Output
<pre> 1 //24K0559 BAZIL-UDDIN-KHAN 2 3 #include <iostream> 4 using namespace std; 5 6 void AddValues(int Aqr[][7], float Firstaverage[]); 7 void Pollutiondays(int Aqr[][7]); 8 void Display(int Aqr[][7]); 9 void MonthlyAqr(int aqr[][28], float Firstaverage[], float lastAverage[], int Aqr[][7]); 10 void Displayreport(int aqr[][28], float Firstaverage[], float lastAverage[]); 11 12 int main() 13 { 14 15 cout << "Welcome to check on air quality index " << endl; 16 int Aqr[4][7] = { 17 {120, 140, 160, 180, 200, 220, 240}, 18 {80, 100, 120, 140, 160, 180, 200}, 19 {60, 70, 80, 90, 100, 110, 120}, 20 {200, 220, 240, 260, 280, 300, 320} 21 }; 22 float Firstaverage[4]; 23 float lastAverage[4]; 24 AddValues(Aqr, Firstaverage); 25 Pollutiondays(Aqr); 26 Display(Aqr); 27 int aqr[4][28] = { 28 {120, 140, 160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220, 240, 120, 140, 29 160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220, 240}, // City 1 30 {80, 100, 120, 140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200, 80, 100, 120, 31 140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200}, // City 2 32 {60, 70, 80, 90, 100, 110, 120, 60, 70, 80, 90, 100, 110, 120, 60, 70, 80, 90, 100, 33 110, 120, 60, 70, 80, 90, 100, 110, 120}, // City 3 34 {200, 220, 240, 260, 280, 300, 320, 200, 220, 240, 260, 280, 300, 320, 200, 220, </pre>	<pre> Welcome to check on air quality index Average for city 1 is 180 Average for city 2 is 140 Average for city 3 is 90 Average for city 4 is 260 Highest Aqr for city 4 that is 320 Day is 3 City is 1 Day is 4 City is 1 Day is 5 City is 1 Day is 6 City is 1 Day is 7 City is 1 Day is 5 City is 2 Day is 6 City is 2 Day is 7 City is 2 Day is 1 City is 4 Day is 2 City is 4 Day is 3 City is 4 Day is 4 City is 4 Day is 5 City is 4 Day is 6 City is 4 </pre>

main.cpp



Shore

Run

Output

```
1 //24K0559 BAZIL-UDDIN-KHAN
2
3 #include <iostream>
4 using namespace std;
5
6 void AddValues(int Aqr[][7],float Firstaverage[]);
7 void Pollutiondays(int Aqr[][7]);
8 void Display(int Aqr[][7]);
9 void MonthlyAqr(int aqr[][28],float Firstaverage[],float lastAverage[],int Aqr[][7]);
10 void Displayreport(int aqr[][28],float Firstaverage[],float lastAverage[]);
11
12 int main()
13 {
14
15     cout << "Welcome to check on air quality index " << endl;
16     int Aqr[4][7]= {
17         {120, 140, 160, 180, 200, 220, 240},
18         {80, 100, 120, 140, 160, 180, 200},
19         {60, 70, 80, 90, 100, 110, 120},
20         {200, 220, 240, 260, 280, 300, 320}
21     };
22     float Firstaverage[4];
23     float lastAverage[4];
24     AddValues(Aqr,Firstaverage);
25     Pollutiondays(Aqr);
26     Display(Aqr);
27     int aqr[4][28] = {
28         {120, 140, 160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220, 240, 120, 140,
29             160, 180, 200, 220, 240, 120, 140, 160, 180, 200, 220, 240}, // City 1
30         {80, 100, 120, 140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200, 80, 100, 120,
31             140, 160, 180, 200, 80, 100, 120, 140, 160, 180, 200}, // City 2
32         {60, 70, 80, 90, 100, 110, 120, 60, 70, 80, 90, 100, 110, 120, 60, 70, 80, 90, 100,
33             110, 120, 60, 70, 80, 90, 100, 110, 120}, // City 3
34         {200, 220, 240, 260, 280, 300, 320, 200, 220, 240, 260, 280, 300, 320, 200, 220,
```

```
Day is 5 City is 4
Day is 6 City is 4
Day is 7 City is 4

City 1
*****
City 2
*****
City 3
*****
City 4
*****
Highest Aqr for city 1 that is 0
Highest Aqr for city in 4 is 320
lowest Aqr for city in 3 is 60
City first week avg 1 Average is 180
City second week avg 1 Average is 180
City third week avg 1 Average is 180
City last week avg 1 Average is 180
City first week avg 2 Average is 140
City second week avg 2 Average is 140
City third week avg 2 Average is 140
City last week avg 2 Average is 140
City first week avg 3 Average is 90
City second week avg 3 Average is 90
City third week avg 3 Average is 90
City last week avg 3 Average is 90
City first week avg 4 Average is 260
City second week avg 4 Average is 260
City third week avg 4 Average is 260
City last week avg 4 Average is 260
```